

Inverse and Inverse-Trig Derivatives

Derivatives of inverse trigonometric functions and of inverse functions

Directions:

- Inverse-trig rules with the chain rule built in:

$$\frac{d}{dx} \arcsin u = \frac{u'}{\sqrt{1-u^2}}, \quad \frac{d}{dx} \arccos u = \frac{-u'}{\sqrt{1-u^2}}, \quad \frac{d}{dx} \arctan u = \frac{u'}{1+u^2}$$

- Inverse-function rule: if $f(a) = b$, then $(f^{-1})'(b) = \frac{1}{f'(a)}$. Find a first, then differentiate f , then take the reciprocal — in that order.
- Leave answers exact. Simplify only what genuinely simplifies.

1. Differentiate: $y = \arcsin(3x)$

Answer: _____

2. Differentiate: $y = \arctan(x^2)$

Answer: _____

3. Let $f(x) = x^3 + 2x + 1$. Note that $f(1) = 4$.

(a) Find $f'(x)$: _____

(b) Find $(f^{-1})'(4)$: _____

4. Differentiate: $y = x \arctan x$ (product rule)

Answer: _____

5. Differentiate: $y = \arccos(2x)$

Answer: _____

6. Let $g(x) = x^5 + 3x - 2$. Note that $g(1) = 2$.

(a) Find $g'(x)$: _____

(b) Find $(g^{-1})'(2)$: _____

7. Differentiate: $y = (\arctan x)^2$ (chain rule with $\arctan x$ as the inside function)

Answer: _____

8. Differentiate: $y = \arctan(\sqrt{x})$ (chain rule; simplify the denominator into a single product)

Answer: _____

9. Differentiate: $y = x^2 \arccos x$ (product rule, with an inverse-trig factor)

Answer: _____

10. Challenge. Let $f(x) = 2x + \sin x$, and note that $f(0) = 0$.

(a) Find $f'(x)$: _____

(b) Find $(f^{-1})'(0)$: _____

(c) Explain why f has an inverse on the whole real line, using your answer to (a).